Fast-Kummer signed recovery gate

KUMMER-RECOVER-20261022

Input: 21 models and the fixed 1,260 source-verified target cohort.

Zero-coordinate xADD: 175/175 quotient-only two-torsion repairs.

Signed target recovery: 1,260/1,260 correct, 0 wrong.

Generic corrected Recover: 1,257; D=0 exceptional lifts: 3.

Literal displayed equations: 1,257 wrong, 3 inapplicable; raw retained.

Eq. (12) G3 coefficient uses -(xi2 Z1+Z2) after reducing

(G3 x-G4) a_P modulo a_Q in the implemented Mumford convention.

Eq. (6) independently supplies b1(Q)^2; G3^2 then follows.

Displayed Eq. (11) differs by 2 f5 xi3 Z2 Z3 in our quintic case.

The three exceptional lifts enumerate field square roots, check

b(x)^2=f(x) mod a(x), and select sign using K(Q+P).

They use Jacobian additions and O(p) field search; not fast.

Closed labeled collector and paired rho pricing: not run.

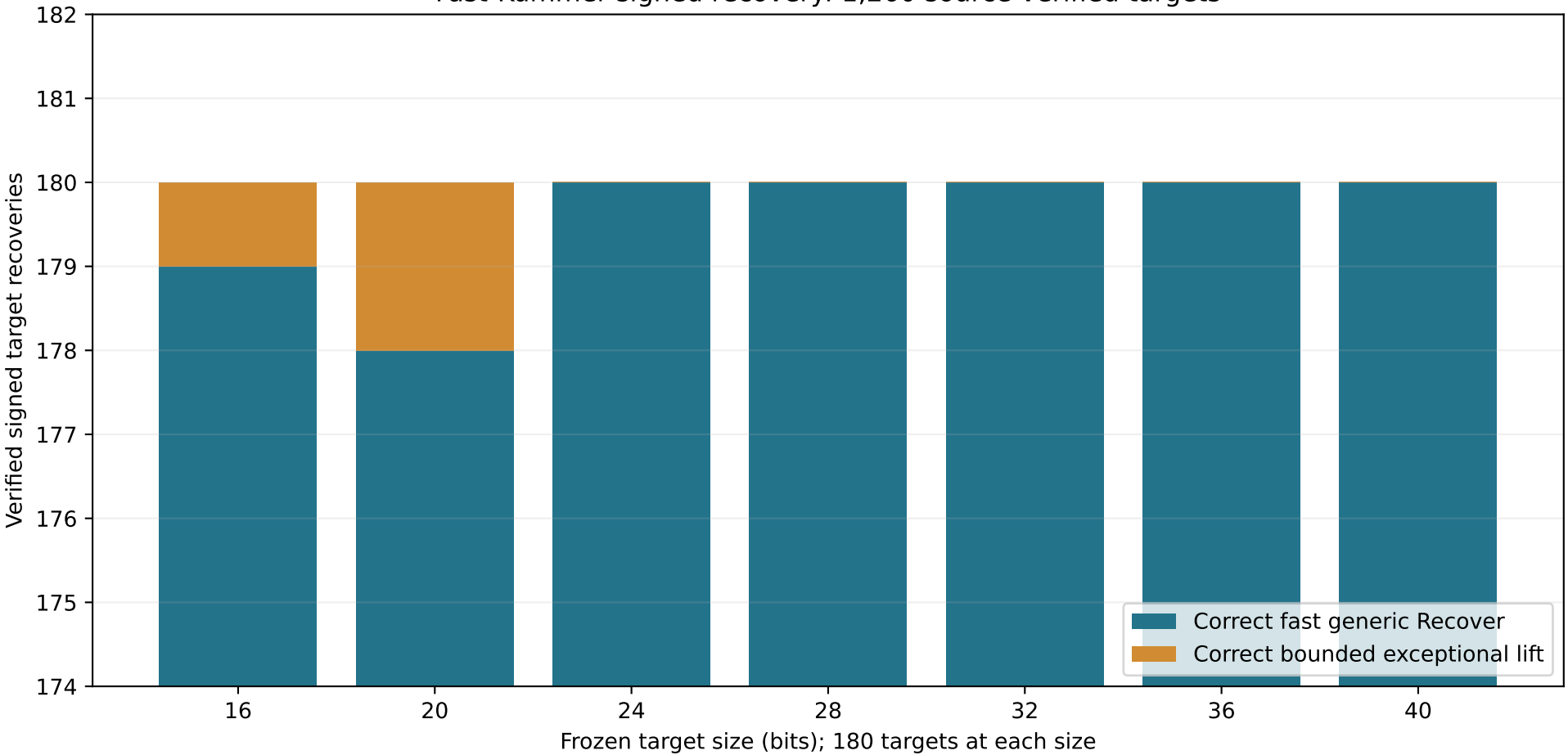
Candidate cost, rho cost and candidate/rho ratio: null.

Rows: results/recover-target-*.jsonl, zero-shift.jsonl,

and results/literal-recover-failures/recover-target-*.jsonl.

Primary source: Chung–Costello–Smith, §§6.4, 7.4

https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/recovery.pdf


Fast-Kummer signed recovery: 1,260 source-verified targets
Verified signed target recoveries
182
181
180
179
178
177
176
175
174
16
20
24
28
32
36
40
Frozen target size (bits); 180 targets at each size
Correct fast generic Recover
Correct bounded exceptional lift